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Lower Cretaceous magnetic stratigraphy in Umbrian pelagic limestone sections

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The magnetic stratigraphy of the Lower Cretaceous, pelagic Maiolica limestone has been investigated in three partially correlative sections at Gorgo a Cerbara, Presale and Frontale in the northern Umbrian Apennines of central Italy. The white, well-bedded limestone has a magnetic mineralogy dominated by magnetite. Stable magnetic directions isolated by thermal demagnetization define alternating polarity zones in each section. The magnetozone patterns are distinctive and can be correlated with the geomagnetic reversal history derived from the M-sequence marine magnetic anomalies. The three new sections confirm the polarity sequence for anomalies M0 to M10N. Although the Maiolica is inadequately dated, the correlated anomalies, together with the results of other investigations, allow tentative associations of anomalies M0–M19 with individual stages in the Lower Cretaceous and Upper Tithonian.

The investigations also demonstrate the usefulness of magnetic stratigraphy in basin analysis. They yield mean sedimentation rates, confirm that there is a hiatus between the base of the Presale section and the underlying Jurassic formations, and show that a large part of the Frontale section has been cut out by faulting.

1. Introduction

Magnetostratigraphic studies in pelagic limestones and deep-sea sediments have confirmed the history of geomagnetic polarity derived from oceanic magnetic anomalies for the Paleogene, and the Late and middle Cretaceous [1–5]. The polarity sequence corresponding to the M-anomalies of the Upper Jurassic and Lower Cretaceous has been confirmed only partially by independent investigation. Channell et al. [6] correlated polarity zones in the Cismon section in the Southern Alps with marine magnetic anomalies M0–M3. Although a much longer, well-defined reversal sequence was obtained, it could not be correlated clearly with the corresponding part of the marine magnetic anomaly sequence.

Parts of the Lower Cretaceous magnetic polarity sequence have been investigated in three Umbrian sections. In the Gorgo a Cerbara section Lowrie et al. [7] correlated M0 with the basal Aptian and associated M1–M3 with the Barremian stage, in agreement with the Cismon results [6]. The Cismon and Gorgo a Cerbara sections were dated with calcareous nannofossils and planktonic foraminifera. The dating of older Southern Tethyan pelagic limestones using nannofossils becomes very difficult because of poor preservation and does not permit resolution of the Hauterivian and Valanginian stages. The Valanginian and Berriasian can be dated paleontologically using the zonation of calpionellids.

Lowrie and Channell [8] studied the location of the Jurassic/Cretaceous boundary in the Umbrian Maiolica limestone in the Bosso section, near Cagli, using the calpionellid biozonation of Micarelli et al. [9]. They correlated anomalies M13–M15 with the Valanginian, M16 and M17 with the

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Berriasian and M18 and M19 with the uppermost Tithonian. The Jurassic/Cretaceous boundary was associated with the base of the negative polarity chron corresponding to M17. However, these Umbrian data differ slightly from results obtained in sections in the Southern Alps by Ogg [10], who located the Jurassic/Cretaceous boundary at the base of the positive polarity chron between M17 and M18 using as a paleontological criterion the rapid increase in abundance of *Nannoconus colomii*.

Magnetostratigraphic results from Deep Sea Drilling Project (DSDP) Site 534 [11] show minor differences in the Jurassic/Cretaceous boundary location depending on whether calpionellids, dinoflagellates or nannofossils were used for dating. At this site the increase in abundance of *N. colomii* was found to correlate with the positive interval between M18 and M19.

The discrepancy between the boundary location in the Bosso section and the results from DSDP and the Southern Alps was thought to arise because Micarelli et al. [9] used a non-standard calpionellid biostratigraphy [12,13]. However, in another Umbrian Maiolica section Cirilli et al. [14], using the standard calpionellid dating scheme for the Western Mediterranean [15], also locate the Jurassic/Cretaceous boundary at a similar level in M17. The difference between the boundary locations in Italian limestones may reflect a small mismatch in the calpionellid and nannofossil dating schemes in the Tethyan realm [8], which does not appear to be the case in the western North Atlantic. The different correlations of the *N. colomii* abundance increase in the Southern Alps and at DSDP site 534 may reflect different timing of these events in the Tethys and North Atlantic oceans [11].

The dating of the Barremian, Hauterivian and Valanginian stages in the Maiolica limestone is the subject of current paleontological research. We have investigated the polarity sequences in three Umbrian sections with the goal of correlating them against each other and to the oceanic magnetic anomaly sequence, and as far as possible placing limits on the ages of these anomalies in the light of available paleontological data. For this study we have expanded the Gorgo a Cerbara section [7]

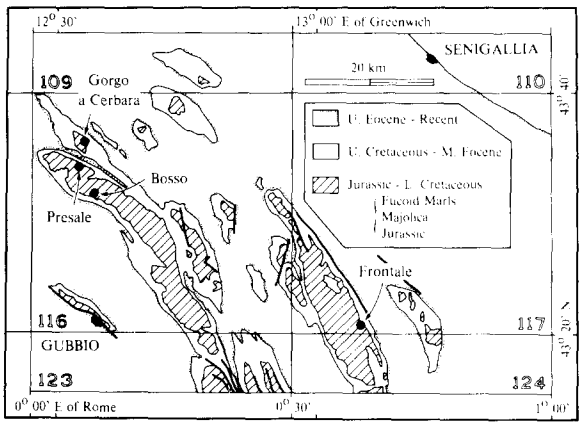


Fig. 1. Locations of the three magnetostratigraphic sections studied in this paper, and the Bosso section of Lowrie and Channell [8], in the Apennines north of Rome and east of Florence. Stratigraphic evidence indicates that a Jurassic/Early Cretaceous listric normal fault, downthrown to the northeast, is hidden beneath the syncline between Presale and Gorgo a Cerbara. Such a fault would explain the striking difference in thickness between these two sections.

and added the Presale and Frontale sections (Fig. 1). In addition to extending the known correlation between magnetostratigraphy in sedimentary sections and the marine magnetic anomaly record, the present work provides useful information on the nature and timing of listric normal faulting during deposition of the Umbrian sequence.

2. Lithology and magnetic character of the Maiolica limestone

The Middle and Upper Jurassic limestones and marls of the Umbrian sequence were deposited in a pelagic environment on the southern margin of the Tethyan ocean. The very cherty, Tithonian "Calcari diasprini" limestone is followed by the Maiolica limestone which occupies the uppermost part of the Tithonian, the lowest four stages of the Cretaceous and much of the Aptian. The Maiolica is a uniform, white, pelagic limestone which contains frequent grey and black chert beds. It is followed stratigraphically by the Fucoid Marls, of late Aptian and Albian age. The lithologic contact is gradational and can be placed for practical

purposes at the highest occurrence of chert nodules in the Maiolica.

The stratigraphic thickness of the Maiolica limestone is quite variable, amounting to 400–500 m in some places but less than 100 m in others. This is the result of active synsedimentary extensional faulting, probably of listric character, related to stretching, thinning, and subsidence of Umbrian crust during growth of the Tethyan ocean [16]. The thick sections often contain slumped intervals, and faults are common. These can usually be recognized during paleomagnetic sampling if the exposure is good, but necessitate careful lithostratigraphic control. The Jurassic pelagic carbonate rocks in the Umbrian Apennines were deposited in a realm of subsiding basins separated by fault-block seamounts [17,18]. The Maiolica was deposited on this uneven submarine topography, filling up the deep basins so that at the end of the Maiolica deposition in the Aptian the submarine topography was largely levelled out [9]. The overlying Fucoid Marls formation has only limited differences in thickness or lithofacies [19,20]. The thinner Maiolica sequences were deposited on the topographic highs and are condensed or reduced, whereas the thicker sections represent the continuous deposition of the basinal facies [9]. The lithostratigraphy of the Umbrian Maiolica is described by Cardellini [21], and the very similar Biancone, or Maiolica of the southern Alps, was studied by Weissert [22].

The Maiolica limestone has a very weak remanent magnetization. Each specimen was measured in at least three positions with a three-axis cryogenic magnetometer and the repeatability of the three determinations was used as a quantitative test of the acceptability of the measurement [23]. The average intensity of the cleaned characteristic remanent magnetization in the 460 samples considered of acceptable quality for this project was 25 μA/m. Samples weaker than about 6 μA/m or having a measurement dispersion larger than about 20 degrees were rejected. About 15% of the sample collection had to be eliminated on these grounds.

Previous work in the Gorgo a Cerbara section [7] was based on samples cleaned magnetically by alternating field (AF) demagnetization. Thermal demagnetization and AF demagnetization of twin specimens from a Maiolica sample generally defined the same stable component of magnetization [8]. However, the Maiolica lithology is not homogeneous. It shows minor but appreciable variation from one geographical locality to another and within the same section. In pilot samples from the sections investigated for this study, progressive thermal demagnetization gave well-defined vectors which decayed uniformly towards the origin of a vector diagram [24] (Fig. 2). Isothermal remanence (IRM) acquisition showed that the magnetic mineralogy of the Maiolica is dominated by a moderately low coercivity component (magnetite), and that a hard component (possibly hematite)

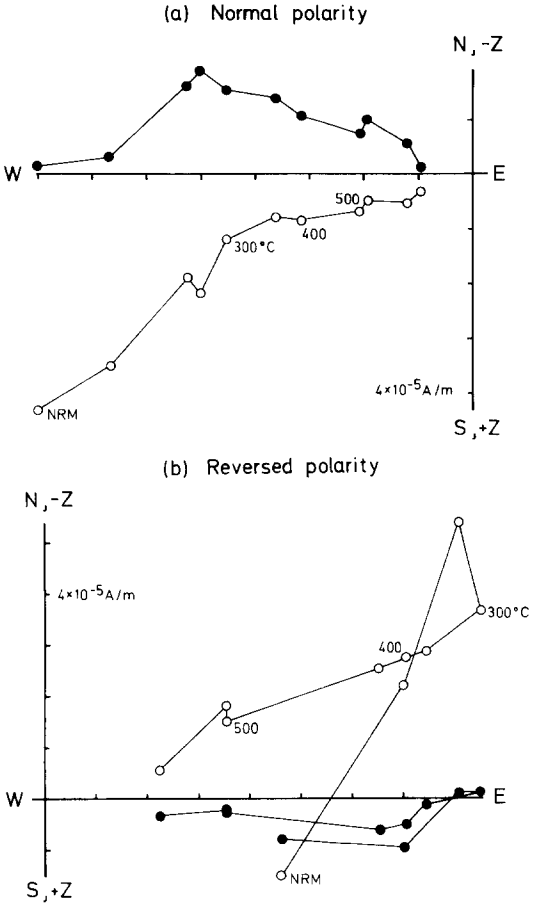


Fig. 2. Thermal demagnetization characteristics of normal and reversed polarity Maiolica samples. Solid points represent projections on the horizontal plane and open points projections on the E-W vertical plane. Numbers on the diagrams give temperature in degrees Celsius.

with coercivities in excess of 0.3 T was also present [7]. Magnetic cleaning of all new samples for this study was carried out thermally, mostly at several temperatures in the range 300–450°C. In the upper part of the Gorgo a Cerbara section the magnetic stratigraphy of Lowrie et al. [7], based on AF demagnetization, was adopted.

3. Location, lithostratigraphy and sampling of the sections

Two of the sections studied, Gorgo a Cerbara and Presale, are only 3.5 km apart, but lie on opposite sides of an abrupt stratigraphic discontinuity. Preliminary stratigraphic evidence suggests that this discontinuity may mark the site of a listric normal fault that was active during Jurassic and Cretaceous sedimentation. The reasons for this inference are: (1) abrupt thickness changes occur at the discontinuity; (2) slide masses up to 200 m across, derived from the side with thin stratigraphy, are intercalated in the adjacent parts of the thick sequence, suggesting the presence of a scarp during deposition; (3) slide masses in the thick sequence at Gorgo a Cerbara moved toward the base of the inferred scarp, suggesting that this sequence was deposited in a tilted half graben, which is characteristic of listric fault situations. The two sections make possible a comparison of the thin, seamount facies on the upthrown block (Presale) with the thick, basinal facies on the downthrown side (Gorgo a Cerbara). The third section, Frontale, is located about 50 km to the southeast, in an area of complex synsedimentary tectonics, where the distinction between fault-block seamounts and basins may not be applicable.

The Gorgo a Cerbara section is located 3.5 km northeast of Piobbico, where excellent exposures occur in the bed of the Candigliano river. The section is on the northeast limb of the Cerbara anticline, with moderate northeasterly dips. Except for a few easily recognized faults, there is little tectonic disturbance. Lowrie et al. [7] determined the magnetic polarity stratigraphy of the lower Fucoid Marls and upper Maiolica in this section. They defined the stratigraphic level of the contact between the two formations (based on the highest occurrence of chert) arbitrarily as 900 m, and the section was sampled down to 842 m. The present study continues the investigation down to 718 m. Below this level the frequency of slumps is so great

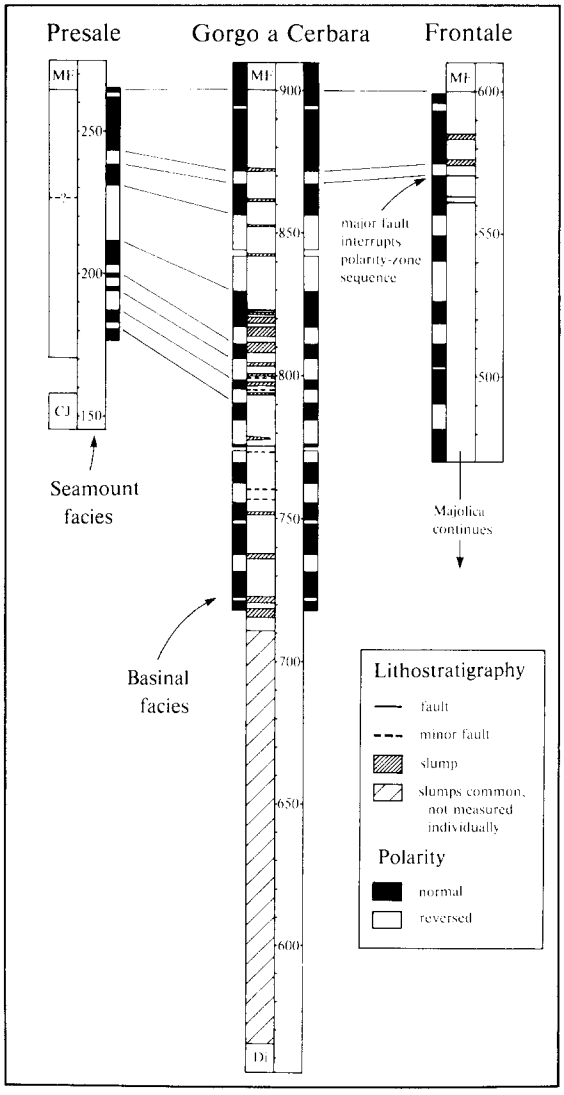


Fig. 3. Stratigraphy and magnetic polarity zonations of the sections studied. The Maiolica is a very fine-grained white limestone with black chert. It is overlain by the Fucoid marls ("Marne a Fucoidi" = MF). In basinal sequences it is underlain by very cherty limestones ("Diaspri" = Di), and in seamount sequences by nodular limestones of the thin "Condensed Jurassic sequence" (CJ). The basinal Gorgo a Cerbara Maiolica is almost four times as thick as the seamount Maiolica of the nearby Presale section. A synsedimentary fault probably lay between them.

that sampling was terminated (Fig. 3). Above 775 m the section is on the southeast side of the river; below that level it is on the northwest side. Although the lowest part of the Maiolica was not sampled, the formation continues down to 565 m, where it rests on the cherty limestones of the "Diaspri". The total thickness of the Maiolica in this section is 335 m, which is about the same as that of other basinal sections in this area (Bosso valley, 9 km to the south-southeast: 323 m; Fosso Giordano, 8 km to the south-southwest: 382 m).

The Presale section is on the northeast flank of the Monte Nerone anticline, along a trail and aqueduct pipe leading to a breached dam in the valley of the Fosso Presale. The trail starts from the end of a new gravel road that turns off the Piobbico–Secchiano road at the cemetery of Rocca Leonella. With the exception of a possible fault at a spring in a thin covered interval, and one easily traced, open fold, the section is undisturbed.

The Monte Nerone anticline roughly coincides with a Jurassic/Cretaceous fault-block seamount, as shown by the fact that most formations of this age are thinner on Monte Nerone than in the nearby basinal areas. This is the case with the Maiolica, which is 94 m thick in the Presale section, compared to 300–400 m in the basinal sections mentioned above. In hand sample, the limestones and cherts of the Presale Maiolica are very similar to those at Gorgo a Cerbara. The principal lithologic difference between the sections is the complete absence of slumps at Presale, which is in agreement with the interpretation that this was a seamount overlooking an adjacent basin.

The Frontale section is located about 2 km west of the village of Frontale, along a road to Monte San Vicino. The top of the section, at the Fucoid Marls/Maiolica contact, was designated 600 m, and sampling was carried down to 470 m. Two substantial faults, whose throws could not be determined, cut the section, and one certain and two possible slumps were noted. This section was intended as a check on the lateral persistence of magnetic polarity zones in the Maiolica and was not studied in the same detail as the other two sections. The tectonic situation in the Frontale–Monte San Vicino section is complex [25,26], reflecting synsedimentary breakup into fault blocks, many with dimensions of only a few hundred meters. As a result, facies types change abruptly and the pattern of faults and facies cannot be clearly resolved with the outcrop density available.

The deposition of the Maiolica limestone (late Tithonian in early Aptian) represents an estimated time interval of about 25–30 Myr [27]. Based on Gorgo a Cerbara (Fig. 3), the typical thickness corrected for slump repetition is about 250 m and gives a maximum sedimentation rate of about 8–10 m/Myr. The sampling interval adopted for the Gorgo a Cerbara and Presale sections was 0.75–1 m, which gives a polarity interval resolution of 75,000–125,000 years. The Frontale section was sampled less intensively. The samples were taken at approximately 1.5-m intervals, and the resolution of polarity intervals is correspondingly less than in the other sections.

4. Magnetic stratigraphy

The stable characteristic remanent magnetization directions in the three sections (Table 1) have fairly large within-site scatter, with angular standard deviations around 20°. Due to the large number of samples from each section the mean directions are very well defined with α_{95} values from 2.5 to 4°. The mean directions differ significantly from section to section, most notably between the western area (Gorgo a Cerbara and Presale) and the eastern area (Frontale section). Similar results from the Upper Cretaceous to Eocene Scaglia limestone were interpreted as evidence for local differential tectonic rotations within the Umbrian Apennines [28]. The rather shallow inclinations reflect the location of the depositional environment of the Maiolica limestone on the African plate on the Southern Tethyan margin. The westerly declinations are evidence for large-scale anticlockwise tectonic rotation of the Umbrian Apennines. This has been variously interpreted as due to a two-phase rotation of an Italian microplate [29–31] or allochthonous displacement of the Umbrian Apennines [28,32].

In magnetostratigraphic sections of the Umbrian Scaglia limestone declinations varied sys-

TABLE 1

Summary of site-mean paleomagnetic directions after simple correction for local bedding tilt

Section	Number of samples	Site mean direction (°)		α_{95} (°)	k	Angular standard deviation
		declination	inclination			
Gorgo a Cerbara	237	275	40	2.5	14.1	21.6
Presale	134	282	41	2.9	18.3	18.9
Frontale	90	297	34	4.1	14.5	21.3

α_{95} is the radius of the circle of 95% confidence, and k is the dispersion parameter. The angular standard deviation is given by $81/k$.

tematically with stratigraphic level but exhibited different trends in each section [28]. To see if the scatter in directions in Maiolica sections is stratigraphically dependent, the data from each section were grouped into a number of "subsites", each consisting of 10–20 samples representing about 10 m of limestone. The mean direction of each subsite was calculated and associated with the stratigraphic level of the mid-point of the subsite. In a section of Scaglia Bianca at Furlo, Alvarez and Lowrie [33] observed rotated declinations in synsedimentary slumped segments some 10 m thick. No evidence for such rotations was found in the Maiolica sections.

Linear regression was used to calculate the slopes of the best-fitting trend lines through the subsite mean declinations and inclinations (Fig. 4). With rare exceptions the trend lines pass through all subsite error bars. Student's t-statistic showed that none of the observed trends was significant at

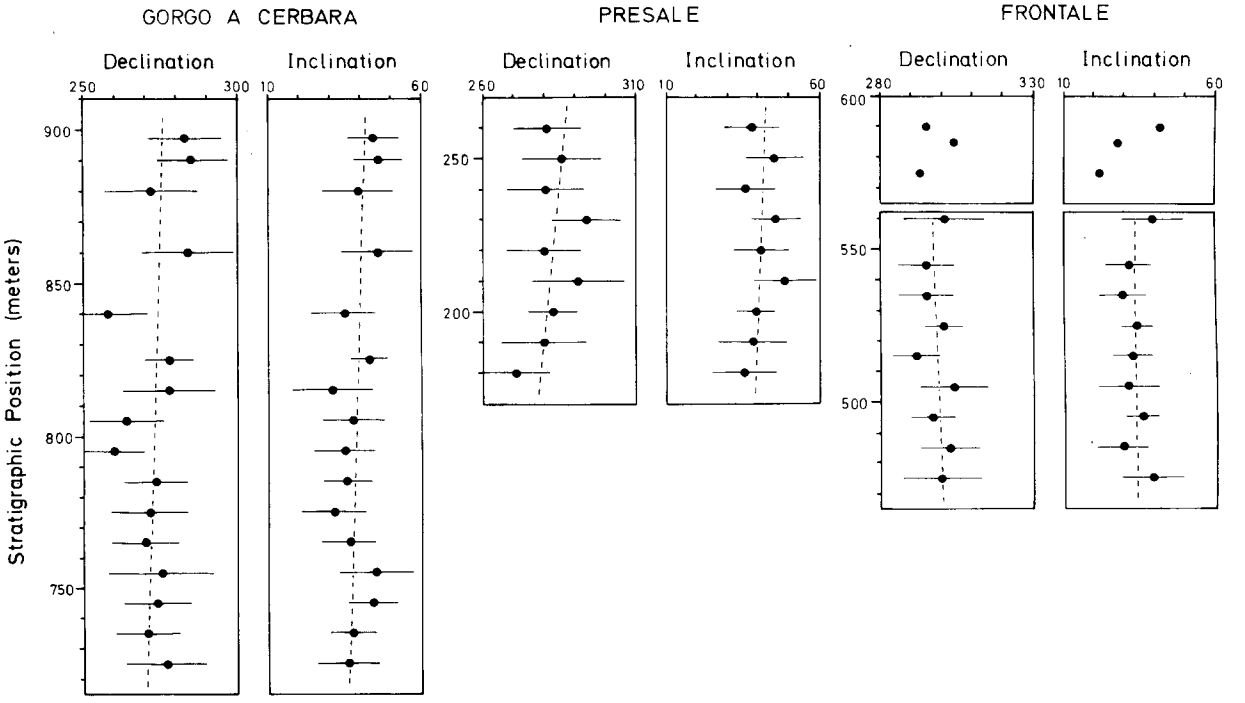


Fig. 4. Analysis of directions grouped stratigraphically to form sites. No declination or inclination trends significant at the 95% confidence level are present in any of the sections.

the 95% confidence level. This implies (a) that no major between-bed tectonic rotation has occurred within the Maiolica sections, and (b) that motion of the depositional site on the African plate during the Early Cretaceous did not have a major northward component relative to the paleomagnetic pole, although east-west motion about the pole is admissible. The latter conclusion is compatible with the reconstructed European and African plate motions during the opening of the Atlantic Ocean [34].

The declinations and inclinations of the bed-

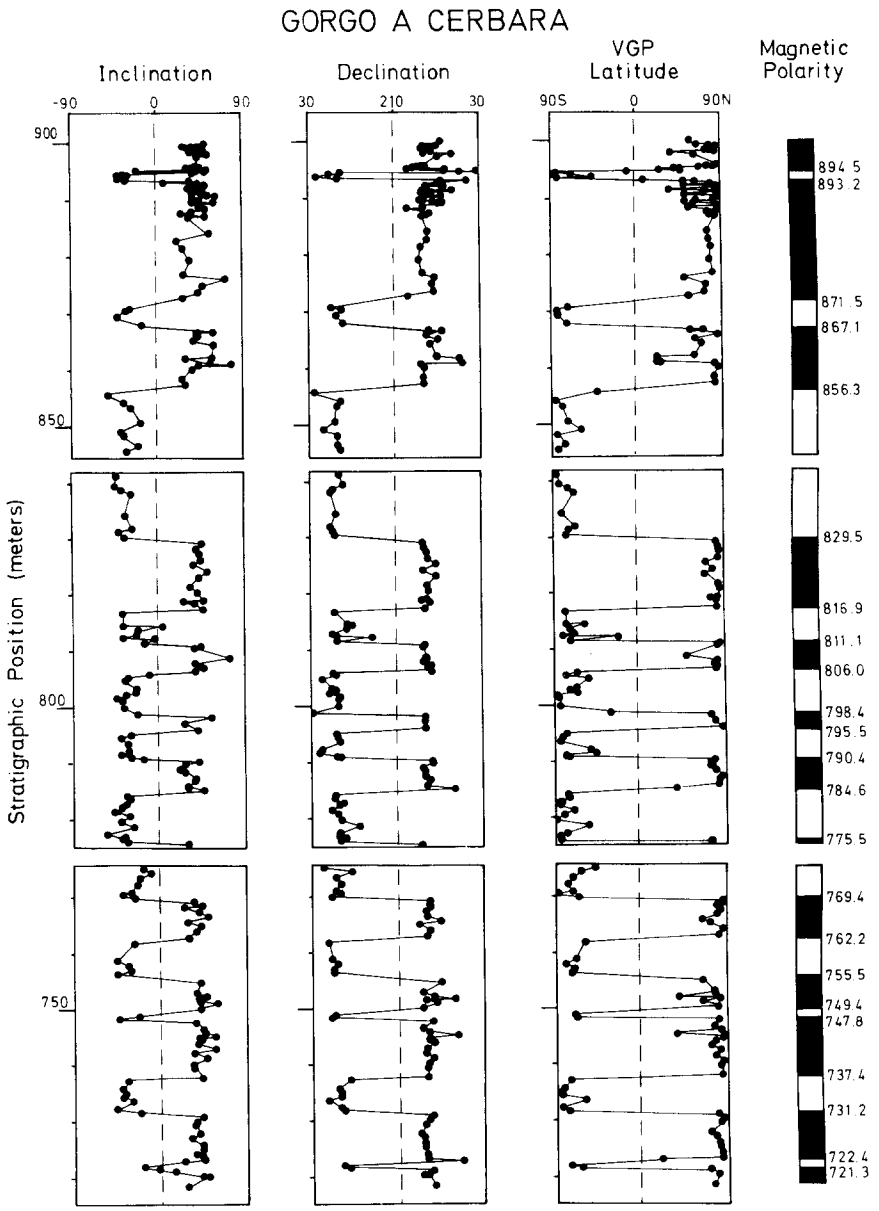


Fig. 5. Stratigraphic variations of the inclination, declination, VGP latitude and polarity in the Gorgo a Cerbara section. Numbers on the polarity column give approximate stratigraphic level of each polarity boundary in meters above an arbitrary base.

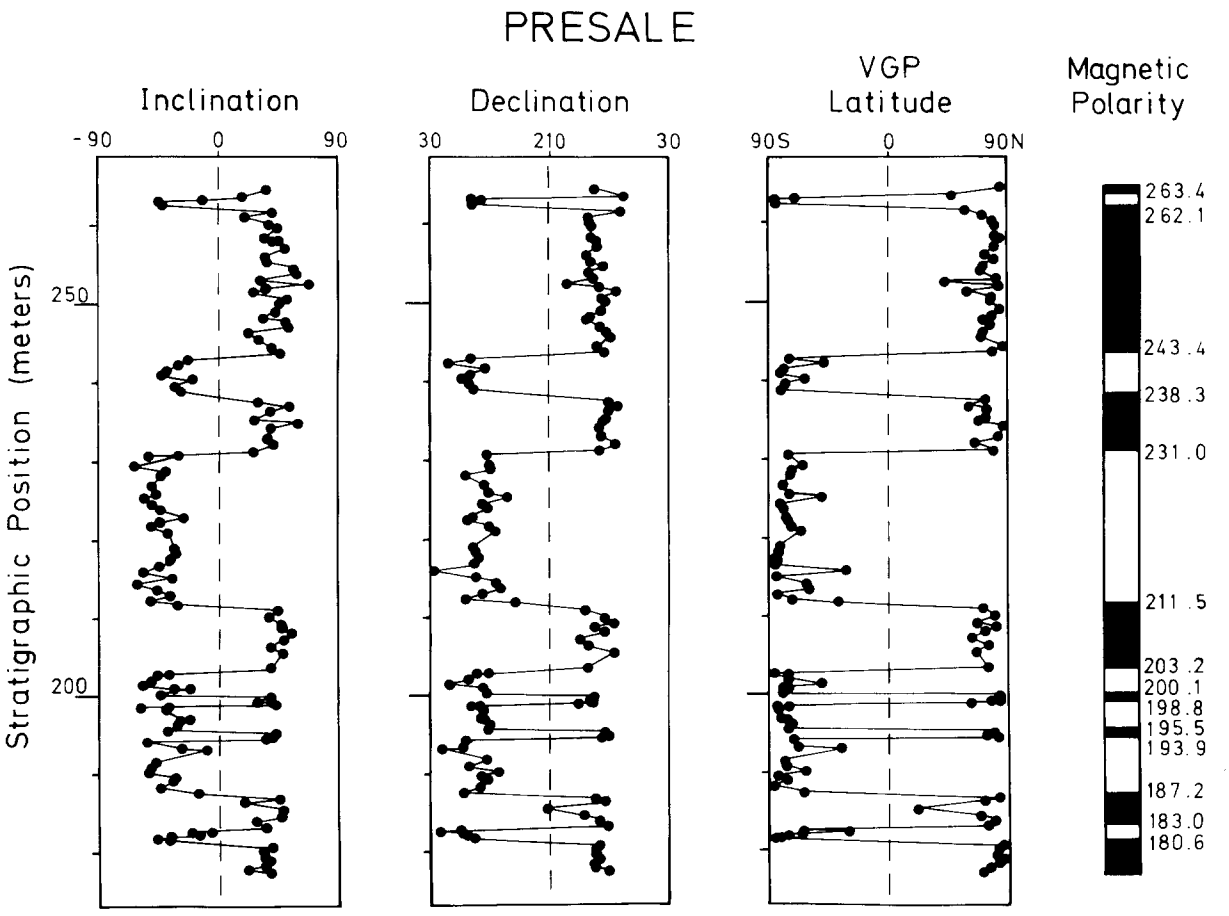


Fig. 6. Stratigraphic variations of the inclination, declination, VGP latitude and polarity in the Presale section.

ding-corrected directions form distinct zones of alternating antiparallel directions to facilitate polarity interpretation, the variation of the latitude of the virtual geomagnetic pole (VGP) during deposition was computed for each section (Figs. 5, 6, 7) in the following way. The paleolatitude of the site during deposition was first calculated from the site mean inclination. The VGP latitude for each sample was then found using this site paleolatitude, the measured inclinations and the declinations corrected for Umbrian tectonic rotation by subtracting the site mean declination. The stratigraphic levels of the reversal boundaries and the thicknesses of the polarity zones were determined from the zero-crossings of the VGP latitude parameter.

The Gorgo a Cerbara section contains slumped intervals, small faults for which the throw could be determined and two major faults of unknown throw (Fig. 3). The section was effectively subdivided into three segments by the faults at 842 m and 775 m. The polarity zones above 840 m were described by Lowrie et al. [7]. The normal zone at the base of that section is rejected here, because it was based on a single poorly defined direction. The polarity above and below the fault at 842 m is negative and we interpret this to be the same polarity zone. It is not possible to estimate how much of the section in this negative polarity zone has been removed or repeated by the fault.

The Presale section was virtually free of disturbances that might have affected the magnetic

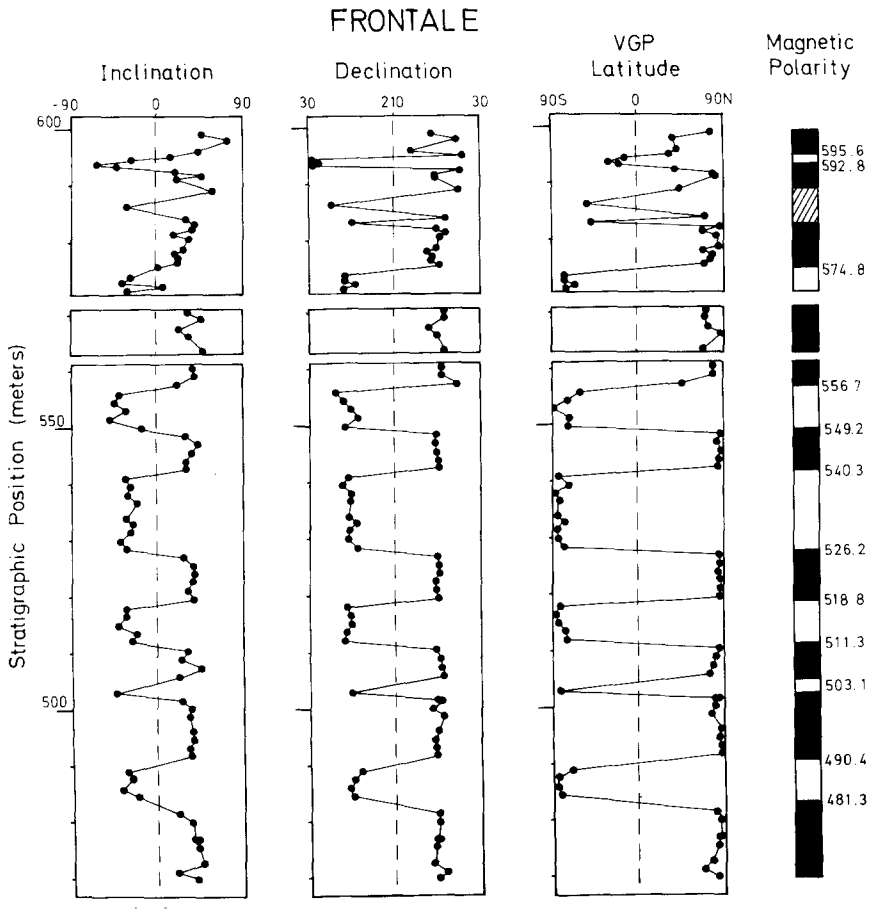


Fig. 7. Stratigraphic variations of the inclination, declination, VGP latitude and polarity in the Frontale section. The cross-hatched zone corresponds to a disturbed part of the section.

stratigraphy. In the Frontale section, however, severe faults at 561 m, 563 m and at 571 m subdivided this section into three segments. The bedding in the top 10 m of the section was wavy. The paleomagnetic vectors are rather irregular, suggesting that the disturbance post-dates fixation of the remanent magnetization direction in the sediment.

5. Magnetic polarity correlations

The standard record of geomagnetic polarity for the Early Cretaceous and Late Jurassic is derived from the M-sequence marine magnetic anomalies [35,36]. The ages associated with these magnetic reversals were interpolated and extrapolated from only two [35] or three [36] calibration points and are probably inaccurate. Both time scales attribute a mid-Aptian age to M0 which is probably basal Aptian in age [6,7,27]. In both studies the Jurassic/Cretaceous boundary was interpolated on the basis of its assumed numeric age. Larson and Hilde [35] placed it just older than M16 and Vogt and Einwich [36] placed it just older than M17. The Larson and Hilde [35] time scale assumes constant spreading for the period of Maiolica deposition and allows direct comparison of the magnetostratigraphic results with the oceanic record.

5.1. *Matching of magnetozones and polarity chrons by cross-correlation*

Identification of a sequence of magnetozones in a section with the corresponding part of the geomagnetic polarity sequence requires the presence of a recognizable pattern in the polarity interval lengths. The patterns can often be correlated visually, especially when the expected age ranges are known. Alternative correlations are sometimes hard to detect by eye but can be detected mathematically by cross-correlation which also allows the significance of the correlation to be tested statistically. This procedure proved effective in analysing Late Miocene magnetostratigraphic sections in Crete [37] and is here applied to the Umbrian Maiolica sections.

The successive thicknesses of the magnetozones in the Maiolica sections were computed in stratigraphic sequence, omitting zones defined only by single samples. The successive durations of polarity chrons in the Early Cretaceous and Late Jurassic polarity time scale [35] were calculated, omitting chrons shorter than 130,000 years, which we take to be the effective magnetostratigraphic resolution. The shorter sequence of magnetozone thicknesses was moved past the sequence of polarity chron durations and the correlation coefficient was computed at every alternate match position (when like polarities are compared). The minimum value of the correlation coefficient which is significant at the 95% confidence level was calculated for comparison. The correlograms (Fig. 8) show all significant matching positions. The optimum correlations (Fig. 9) were selected from the significant ones on the basis of available age information for each section. It must be emphasised that cross-correlation of geological sequences strongly depends on constancy of the rate processes involved. Pelagic sedimentation in the Tethyan realm and sea-floor spreading appear to satisfy this condition adequately.

The magnetostratigraphy at Gorgo a Cerbara above the fault at 775 m gives 3 significant correlations with the M-anomaly polarity sequence (Fig. 8). One of these is with Jurassic anomalies M21–M25 and can be eliminated. The stronger of the remaining two is with M0–M9 and is our preferred correlation (Fig. 9) because of the known continuity between the Maiolica and the overlying Fucoid Marls in this section and because of microfossil ages from our previous study [7]. The part below 775 m gives two significant correlations of which we favor the stronger correlation with M9–M10N (Fig. 9). The fault at 775 m apparently repeats about 5.6 m of the section.

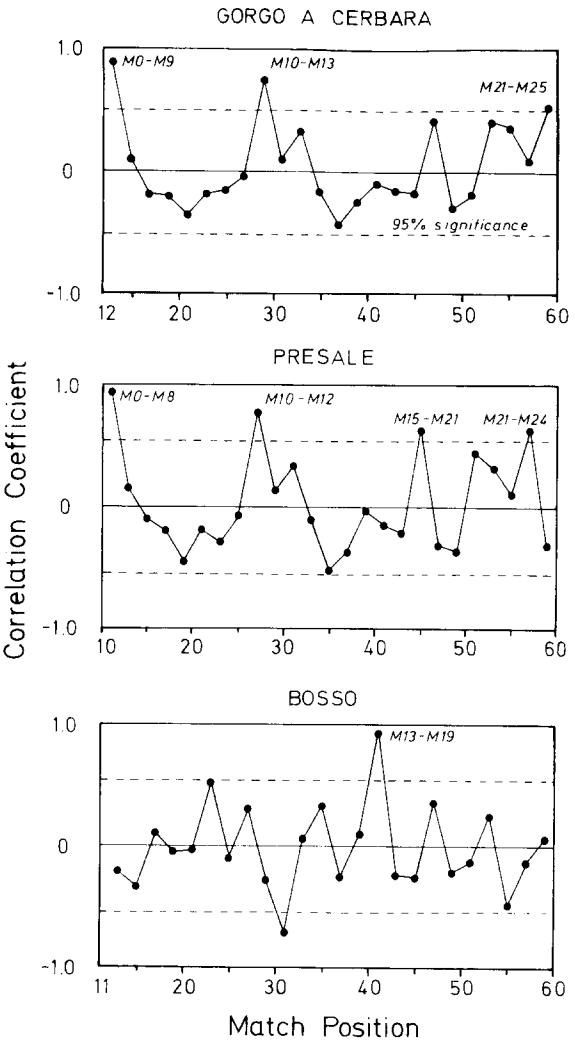


Fig. 8. Cross-correlation profiles between the sequences of magnetozone thicknesses in three Maiolica sections and the sequence of M-anomaly polarity chron durations. Only points lying above the dashed 95% confidence limits represent significant correlations. They are identified with the range of corresponding M-anomaly polarity chrons.

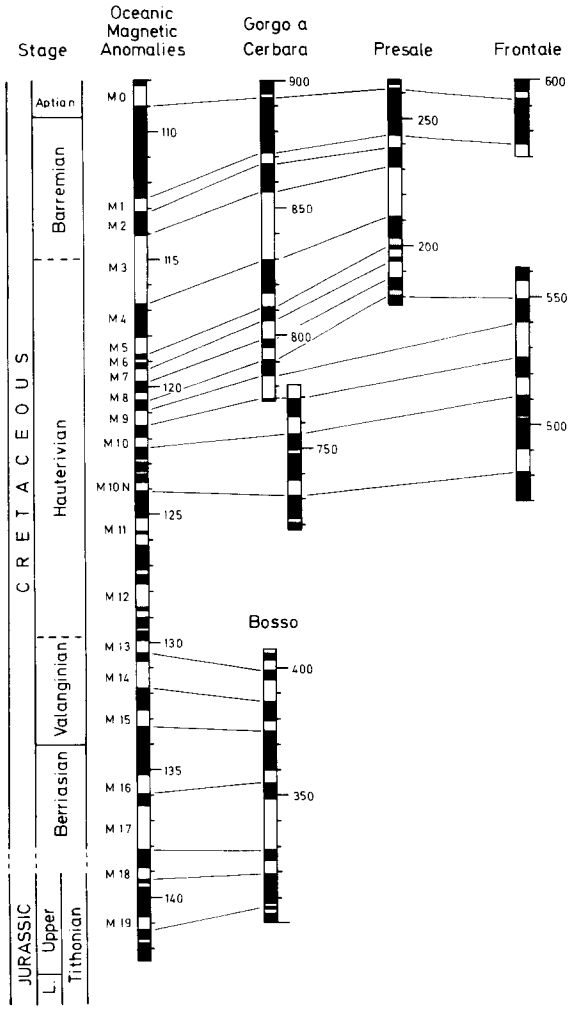


Fig. 9. Correlations between Italian magnetostratigraphic sections and the geomagnetic polarity record derived for the M-sequence marine magnetic anomalies. The anomaly numbers and associated ages in Myr are taken from Larson and Hilde [35]. The Gorgo a Cerbara, Presale and Frontale sections are from the present study, the Bosso section is from Lowrie and Channell [8]. The ages in the left-hand column are based on the paleontological dating of these sections. The polarity column scales give stratigraphic level in meters.

The Presale section gives 4 significant correlations with the oceanic record (Fig. 8). Two Jurassic correlations can be eliminated; the stronger of the remaining two corresponds to marine magnetic anomalies M0–M8 (Fig. 9). The similarity between the cross-correlation profiles of the Gorgo a Cerbara and Presale sections against the M-anomaly sequence is striking (Fig. 8) and demonstrates how well these two sections also correlate with each other (Fig. 9).

The top two parts of the Frontale section are not sufficiently long or distinctive enough to give a unique correlation with the marine record. In spite of the wavy, disturbed bedding it seems reasonable to correlate the topmost part with M0–M1. The bottom part of the Frontale section, below 561 m, correlates best with marine magnetic anomalies M8–M10N (Fig. 9). This correlation implies that the faults cut out about 140 m (7 Myr) of record in the Frontale section. A less significant correlation was found with M13–M16.

Lowrie and Channell [8] investigated the magnetic stratigraphy of the Bosso section (Fig. 1). The Bosso results give only a single, highly significant correlation with the M-sequence (Fig. 8) and match anomalies M13–M19.

5.2. *Estimated average sedimentation rates*

Although the numeric ages of the M-sequence reversals are probably inaccurate, the available time scales [35,36] suffice for using the magnetostratigraphic correlations to calculate mean sedimentation rates. As indicated by the slopes of the tie-lines in Fig. 9, the sedimentation rate decreases upwards in the Gorgo a Cerbara section, with mean values of 9 m/Myr (for M0–M9) and 14 m/Myr (for M9–M10N) above and below the fault at 775m, respectively. The interpreted sedimentation rate for the Presale section is 7 m/Myr for the M0–M8 interval. This is only slightly less than the corresponding rate at Gorgo a Cerbara.

The correlation of the Frontale section below the faulted and disturbed upper part with M8–M10N (Fig. 9) gives a high average sedimentation rate of about 19 m/Myr.

The sedimentation rate derived for the Early Cretaceous and Upper Tithonian (M13–M19) in the Bosso section is 9 m/Myr. Higher in the Bosso section the magnetization intensity of samples which might have correlated with anomalies M10N–M13 was too weak to be used. This might be a result of dilution of the magnetic fraction due to high sedimentation rate or lack of detrital input at that time.

6. Discussion

The Presale section is located on the northern flank of a Jurassic seamount (Monte Nerone) and is immediately underlain by Jurassic pelagic rocks. It is considered to be a condensed or reduced section of the typical seamount facies. Despite the difficulty of paleontological dating, biostratigraphic studies have indicated the presence of a hiatus at the base of the Monte Nerone seamount facies Maiolica [9,21].

This is confirmed by the present magnetostratigraphic study. The magnetostratigraphic correlations (Figs. 3, 9) show that the Presale sedimentation rate (7 m/Myr) is only slightly less than the contemporary rate at Gorgo a Cerbara (9 m/Myr). The Presale section is not complete but covers only the lower Aptian, the Barremian and the upper part of the Hauterivian. The polarity sequence corresponding to the lower Hauterivian, the Valanginian, Barremian, and upper Tithonian is missing. The paleomagnetic data confirm the presence of a large hiatus at the base of the Presale section.

The magnetozone correlation between Presale and Gorgo a Cerbara clarifies the timing of displacements on the listric normal fault that is inferred to lie between them, and places constraints on the source of sediment that accumulated in the basinal area. Our interpretation is that, from middle Tithonian to middle Hauterivian time, all sediment deposited in this region reached the basinal areas, either directly through sedimentation or by lateral transport, mostly by slumping, from the seamount areas to the basins. By the middle Hauterivian the basinal area had been largely filled in and the irregular submarine topography sufficiently leveled that sediment could also accumulate over the former seamounts, although in slightly lesser thicknesses than over the former basin.

Fig. 3 shows that, although slump accumulation at Gorgo a Cerbara nearly ceases in the upper part of the section, there are still substantial slump masses as high as 825 m, despite the fact that preservation of pelagic sediment had begun somewhat earlier at Presale. This shows that the slump material in at least this upper part of the Gorgo a Cerbara section did not come from the adjacent seamount, or at least not from the exact area of the Presale section. We conclude that the slumps that accumulated at Gorgo a Cerbara probably moved southwestward down the gentle slope toward the foot of the inferred listric fault. This is supported by the southwestward overturn of all four of the well preserved slump folds in the interval in question (785–822 m), as well as the two slump folds observed lower down (717 m, 752 m). Sediment which slumped from the Presale sector of the seamount in early Maiolica time must have accumulated at the base of the fault scarp and not been able to ascend the southwest-sloping side of the basin as far as the Gorgo a Cerbara locality. Although preliminary, these inferences confirm the potential usefulness of paleomagnetic stratigraphy in investigating problems in basin analysis [33].

The Umbrian Maiolica magnetostratigraphic sections cover most of the time interval from the upper Tithonian to the early Aptian, represented by marine magnetic anomalies M0–M19 (Fig. 8). Only the segment corresponding to M11–M13 has not yet been confirmed independently in magnetic stratigraphy investigations.

The results of paleontological dating give the locations of major stage boundaries in the reversal sequence. The Aptian/Barremian boundary was found just below M0 in the sections at Gorgo a Cerbara [7] and Cismon [6]. The position of the Barremian/Hauterivian boundary in the Cismon section was determined (F. Medizza, in [6]) but the magnetizations in this part of the section did not give a clear polarity stratigraphy. The Cismon results imply, however, that the Barremian/Hauterivian boundary lies in or near M3, at the approximate position indicated in Fig. 9.

Paleontological dating of the Bosso section using calpionellids gave the positions of the Hauterivian/Valanginian, Valanginian/Berriasian and Berriasian/Tithonian stage boundaries [9]. The quality of the magnetic data in the Bosso section deteriorated upwards and at the level where the Hauterivian/Valanginian boundary was located the identification of M13 was only tentative. The Valanginian/Berriasian boundary fell in the positive chron between M15 and M16. At DSDP Site

534 this boundary coincided with a strong seismic reflector and a possible hiatus was suspected; the location was also in a positive chron corresponding to M15–M16 or younger. The Berriasian/Tithonian (Cretaceous/Jurassic) boundary was correlated near the old side of negative polarity chron M17 in the Bosso section [8] and in the nearby Fosso del Giordano section [14], near the young edge of M18 in the Southern Alps [10], and in the positive interval older than M18 at DSDP Site 534 [12]. Until this boundary location is better established we assume it to be around M18 (Fig. 9).

The M-anomalies M0-M19 can therefore be dated tentatively as follows:

M0	early Aptian
M1–M3 (upper part)	Barremian
M3 (lower part)–M12	Hauterivian
M13–M15	Valanginian
M16–M17	Berriasian
M18	Jurassic/Cretaceous boundary (approx.)
M19	late Tithonian

From the mid-Triassic Anisian/Ladinian boundary to the mid-Cretaceous Aptian/Albian boundary radiometric control of age boundaries is not adequate to allow satisfactory geologic time scale tie-points. Harland et al. [27] adopted the hypothesis of equal average duration for the Jurassic and Early Cretaceous ages to interpolate the age boundaries between the stated tie points. If the stage boundaries used here are approximately correctly located and if the Umbrian sedimentation rates and the oceanic spreading rates are approximately constant, the Hauterivian appears to have lasted 2–3 times as long as the other Neocomian stages (Berriasian, Valanginian, Barremian). This demonstrates the need for further work on establishing the absolute ages of the stages in this part of the geologic time scale.

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